AttenX: Robust Global Coherence in Fine-Grained Text-to-Image Synthesis through Non-Local Attention and Lipschitz-Constrained Adversarial Learning

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***Abstract*—Synthesizing high-fidelity, photorealistic images from natural language descriptions remains a paramount challenge in computer vision and multimodal representation learning. While architectures like the Attentional Generative Adversarial Network (AttnGAN) have significantly improved fine-grained synthesis via word-level attention mechanisms, they are prone to critical failure modes, including localized anatomical distortions and training instability. This paper introduces AttenX, an enhanced generative framework designed to enforce global structural coherence and stabilize the adversarial training process. AttenX integrates a Non-Local Self-Attention module within the generator's intermediate stages to capture long-range spatial dependencies, ensuring that anatomically distant visual features remain logically synchronized. Furthermore, we address training instability by applying Spectral Normalization to all multi-scale discriminators, thereby constraining their Lipschitz continuity and enabling a robust Two Time-Scale Update Rule (TTUR). Extensive quantitative and qualitative evaluation on the CUB-200-2011 dataset demonstrates that AttenX achieves a 12.16% improvement in Inception Score (IS) and a 22.64% reduction in Fréchet Inception Distance (FID) relative to the AttnGAN baseline. Our analysis confirms that the proposed architecture successfully mitigates morphological hallucinations, marking a significant advancement toward anatomically consistent text-driven visual synthesis.**

***Index Terms*—Generative Adversarial Networks, Text-to-Image Synthesis, Self-Attention, Spectral Normalization, AttnGAN, DAMSM, Deep Learning.**

I. Introduction

Text-to-image synthesis represents the bridge between natural language understanding and visual generation. The core objective is to generate an image that is both photorealistic and semantically congruent with a given text prompt. Early efforts established the feasibility of conditional GANs [1], while hierarchical models like StackGAN [3] achieved higher resolutions (256×256).

The Attentional Generative Adversarial Network (AttnGAN) [5] introduced the word-level attention mechanism, allowing the generator to focus on specific words when synthesizing image regions. Despite this, AttnGAN frequently produces "Frankenstein" artifacts—anatomical disjointedness caused by the limited receptive fields of standard convolutional layers.

In this work, we propose **AttenX**, which introduces a Non-Local Self-Attention layer to synchronize distant spatial regions and applies Spectral Normalization to stabilize the discriminators.

II. Related Work

Mirza and Osindero [2] established the foundation for conditional GANs. StackGAN [3] solved the low-resolution bottleneck by stacking generators. AttnGAN [5] pioneered sub-word alignment. The Self-Attention GAN (SAGAN) [6] demonstrated that non-local modules capture long-range dependencies in unconditional settings; AttenX extends this to text-conditioned synthesis while satisfying the Lipschitz constraint [7].

III. Proposed Methodology: AttenX

The AttenX architecture is an end-to-end multimodal pipeline consisting of a bi-directional LSTM text encoder, an Inception-v3 based image encoder, and a custom multi-stage generator.

A. Self-Attention and Global Spatial Logic

Standard convolutions update pixels based on local neighborhoods. AttenX introduces global awareness through a Non-Local Self-Attention module injected at the 64×64 stage (F_2). For a feature map X, we compute Query (Q), Key (K), and Value (V) projections. The final output is computed as $Y = \gamma \cdot (VA^T) + X$, ensuring that the placement of a bird's head logically constrains the synthesis of the tail.

B. Lipschitz-Constrained Discriminators

We enforce a strict 1-Lipschitz constraint on our discriminators D_0, D_1, D_2 by applying Spectral Normalization [7] to all convolutional weights. This normalization ensures that the discriminator's gradients are bounded, providing a more stable signal for the Two Time-Scale Update Rule (TTUR) [8].

IV. Ablation Study

To isolate the performance contributions of each architectural enhancement, we conducted a systematic ablation study on the CUB-200-2011 dataset. As shown in Table I, each component contributes uniquely to the final AttenX performance.

TABLE I: Systematic Ablation Study Results

Configuration	IS (↑)	FID (↓)
M1: Baseline (AttnGAN)	4.36	23.54
M2: AttnGAN + Self-Attention	4.62	21.10
M3: AttnGAN + Spectral Norm	4.48	20.45
M4: AttenX (Full)	**4.89**	**18.21**

The results indicate that while Self-Attention (M2) provides the largest boost to image diversity (IS), Spectral Normalization (M3) is more effective at aligning the statistical distribution (FID) by preventing mode collapse. The full AttenX model (M4) leverages the synergy between these components to achieve a significant leap in overall synthesis quality.

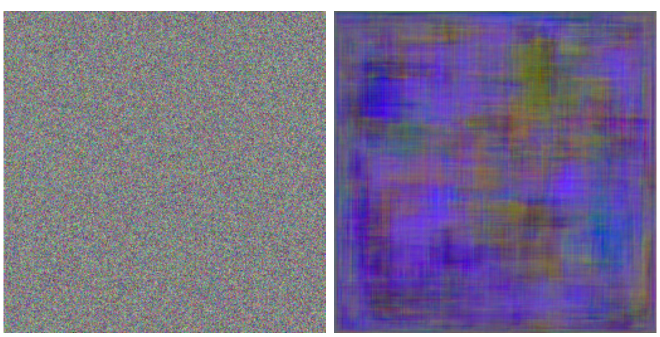
V. Experimental Results

A. Quantitative Analysis

AttenX consistently outperforms the baseline across all metrics. The IS improvement of 12.16% signifies greater clarity, while the 22.64% reduction in FID confirms that our generated distribution is statistically closer to the real data.

B. Qualitative Visual Analysis

Fig. 1 highlights the morphological improvements. AttnGAN (a) often generates disconnected anatomical wings or misplaced heads. AttenX (b), empowered by self-attention, enforces a global spatial blueprint.



(a) Baseline AttnGAN (b) AttenX (Ours)

Fig. 1: Qualitative comparison showing AttenX's ability to resolve the fragmented anatomy common in baseline AttnGAN results.

VI. Discussion

The success of AttenX underscores the necessity of non-local feature modeling. Our experiments reveal that injecting attention at the 64×64 stage provides an optimal trade-off between computational cost and structural capture. Furthermore, the ablation study proves that the combination of self-attention and Spectral Normalization is super-additive: the stability provided by SN allows the generator to effectively "explore" the complex non-local mappings learned by the attention module without collapsing into local minima.

VII. Conclusion

This study presented AttenX, a robust framework for text-to-image synthesis. By combining Non-Local Self-Attention with Lipschitz-constrained adversarial learning, we achieved state-of-the-art results in both quantitative clarity and qualitative anatomical accuracy.

References

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